

DANIEL BENJAMIN

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available starting May 2027 for a 4-month or 8-month co-op position

EDUCATION

University of British Columbia
Bachelor of Applied Science – Mechanical Engineering

Expected Graduation: May 2028

ROBOTICS & SOFTWARE DEVELOPMENT HIGHLIGHTS

- Designed a [robotic hand](#) driven by soft pneumatic actuators 25% better at grasping than previous iterations
- Developed AutoLISP automation scripts to reducing CAD drafting time by over 30%, saving 7+ hours per week
- Deployed SVN server on a VPS to track design changes alongside a [custom SOLIDWORKS integration](#)
- Designed and manufactured an [FSAE chassis frame](#) in SOLIDWORKS and developed Python scripts to calculate forces under various braking and cornering load cases for torsional rigidity simulations in ANSYS Mechanical

SELECTED EXPERIENCE

Chassis Team Lead (prev. Team Member), UBC Formula Electric Sep 2023 – Present

- Led 7-person chassis subteam using Agile project management with Jira and Gantt charts to coordinate deliverables and keep the 2026 competition vehicle on schedule
- Planned and executed carbon fiber + glass fiber composite layups for car body panels and aero kit
- Coded a [custom 3D printer dashboard and Slack bot](#) in Python to manage print jobs, status and reservations
- Wrote detailed design documentation and SOPs to ensure knowledge transfer for future team members
- Designed and validated multiple waterproof [3D-printed enclosures](#) for circuit boards against IPX5 standards

Mechanical Co-op, Avalon Mechanical May 2025 – Aug 2025

- Produced HVAC, plumbing, and fire suppression layouts in AutoCAD for commercial and residential projects
- Assisted with drawing revisions, redlines, and documentation for tender and construction packages
- Performed hallway pressurization analysis and fan sizing using Excel and TRACE, calculating required airflow and pressure differentials to meet NFPA and provincial code requirements
- Validated fire sprinkler hydraulic systems using Elite Software FIRE for code-compliant designs

Undergraduate Research Assistant, UBC MEMS Lab May 2024 – Feb 2025

- Analyzed 15+ tendon-driven prosthetic hands to identify common successful design elements
- Rapidly prototyped design iterations using a custom-built multi-extrusion 3D printer running RepRapFirmware
- Designed a custom silicone return system around the actuators' force limits.
- Developed a demo platform for the hand to autonomously mimic hand gestures using MATLAB and Arduino

Volunteer, Victoria Hand Project July 2022 – Present

- Assembled and repaired 3D-printed low-cost prosthetic arms with voluntary open/close functionality
- Contributed concrete ideas to improve assembly speeds, leading to creation of new documentation

PROJECTS

WeBWorKer Chrome Extension github.com/danielrbenjamin/WeBWorKer

- Published an [extension](#) on the [Chrome Web Store](#) with over 900 users at peak adoption
- Used JavaScript and CSS to provide real-time LaTeX formatted previews of plaintext math on the WeBWorK homework platform, show matching parentheses, recognize variable names, and add custom query integrations

AI-Driven Predictive Maintenance System github.com/danielrbenjamin/manu465-project

- Applied [ML in Python](#) using TensorFlow to predict maintenance needs in smart manufacturing equipment

SKILLS

Mechanical: SOLIDWORKS, AutoCAD, ANSYS, Composites, Machining, 3D Printing, MIG Welding, FMEA

Software: C, MATLAB, Python, AutoLISP, JS, Arduino, Git, SVN, Docker, AWS, Jira, Bluebeam, Adobe CC